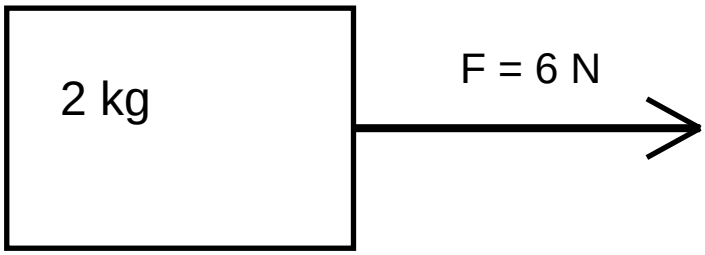
Question 1: Newton's second law

m = 2 kg, F = 6 N.

F = ma

a = F / m = 3 m/s^2

The solution continues on page 2.



Question 1 (continued)

$v = u + at$

$u = 0, t = 4\ s$

$v = 0 + 3 \times 4 = 12\ m/s$

The force points to the right.



motion to the right

Question 2: Energy and an inclined plane

$E_k = (1/2)\, m\, v^2$

$m = 2$ kg, $v = 3$ m/s

$E_k = 9$ J

